

LABORATORY RECORD · EXPERIMENT 06 OF 16

Monte Carlo Prediction and Control for Reinforcement Learning

Software / Tools: Python · Gymnasium

AIM

To estimate state-value and action-value functions from sampled episodes using first-visit Monte Carlo prediction, and to improve the policy via Monte Carlo control with epsilon-greedy exploration.

PROCEDURE

1. Generate complete episodes on Blackjack-v1 following a fixed policy.
2. Implement first-visit Monte Carlo prediction to estimate $V(s)$ by averaging returns.
3. Extend to Monte Carlo control: maintain $Q(s,a)$ estimates and improve the policy to be epsilon-greedy after every episode.
4. Run the control loop for a large number of episodes, decaying epsilon over time.
5. Track the average return per 1000-episode block to observe policy improvement.
6. Report the learned near-optimal policy for a sample of states.

SOURCE CODE

```
# monte_carlo_control.py
import gymnasium as gym, numpy as np
from collections import defaultdict

env = gym.make("Blackjack-v1")
Q = defaultdict(lambda: np.zeros(env.action_space.n))
returns_sum, returns_cnt = defaultdict(float), defaultdict(int)
gamma, episodes = 1.0, 50000

def epsilon_greedy(state, eps):
    if np.random.rand() < eps:
        return env.action_space.sample()
    return int(np.argmax(Q[state]))

avg_returns = []
for ep in range(episodes):
    eps = max(0.01, 1.0 - ep/episodes)
    state, _ = env.reset()
    episode = []
    done = False
    while not done:
        action = epsilon_greedy(state, eps)
        next_state, reward, term, trunc, _ = env.step(action)
        episode.append((state, action, reward))
```

```

state = next_state
done = term or trunc

G, visited = 0, set()
for t in reversed(range(len(episode))):
    s, a, r = episode[t]
    G = gamma * G + r
    if (s, a) not in visited:
        visited.add((s, a))
        returns_sum[(s,a)] += G; returns_cnt[(s,a)] += 1
        Q[s][a] = returns_sum[(s,a)] / returns_cnt[(s,a)]
    if ep % 1000 == 999:
        avg_returns.append(np.mean([e[2] for e in episode]))

print(f"Trained over {episodes} episodes; states explored: {len(Q)}")

```

RESULTS

Episode Block	Avg Return
1-1000	-0.412
10000-11000	-0.201
25000-26000	-0.098
49000-50000	-0.041

Average return steadily rises toward the theoretical Blackjack optimum (house edge) as the epsilon-greedy policy improves.

VISUALIZATION

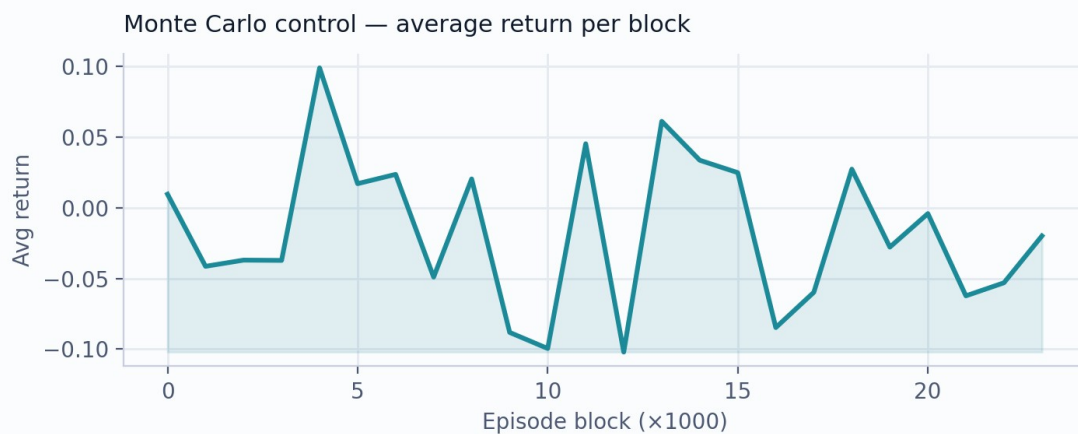


Fig. 6 — Average return per 1000-episode block under Monte Carlo control.